

Phonons for Thermal Safety of Lithium Ion Batteries

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(Dated: 19 April 2026)

Phonon engineering is an attractive method to mitigate the onset, critical, and maximum temperatures of thermal runaway in lithium ion batteries. In this literature review, an ionic solute and thin nanopore matrix are explored to reduce phonon scattering in solid state electrolyte polymers to improve thermal transport. Electron-Phonon interaction is explored in $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ cathode materials, where the relation between Ni content and thermal conductivity, electrical conductivity, ion transport, and current density is shown. This abstract raises several question which I think the rest of the paper does a good job answering for the most part: What is Phonon Engineering? What is thermal runaway? Why should I care about thermal transport in a battery when charge transport is the desired quality?

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I. INTRODUCTION

Lithium ion batteries are a common consumer product for power storage, Does grid storage count as a consumer product? from electric vehicles to grid storage and portable electronics. However, lithium ion batteries experience both joule heating and residual heat from exothermic reactions. Here it would be nice for you to specify the exothermic Chemical reactions that Lithium-Ion batteries are prone to. As the battery heats up, the activation energy for chemical reactions is reduced The fact that the activation energy is reduced seems incredibly strange to me. I would have thought that the reactants would be at an increased energy due to temperature and thus the reaction is more likely even while the height of the energy barrier maintains the same. Maybe have a word or two on why this happens? Or at the very least, cite a paper that explains this please.— leading to a positive feedback loop which poses significant safety risk towards consumer usage (battery fire and explosion). Specifically, this phenomena of thermal runaway occurs when the heat is not dissipated fast enough across the battery, and a region passes a critical temperature. In this literature review, phonon-electron coupling and phonon engineering At this point I would recommend explicitly defining phonon engineering as adding material to a solid in order to alter the properties of the phonons or whatever. I'm assuming that's what that is, but it's best to be explicit or to not use the term at all. are explored to improve I don't think phonon-electron coupling is ever fully explained in your paper. I think you mention that ions disrupt the lattice and will scatter phonons and I believe you said that that was a detriment to the thermal transport, not an improvement. thermal transport in both the cathode and the ion transport membrane of lithium ion batteries to mitigate thermal runaway. Further, changing the vibrational densities Did you mean density of states? and energies of phonons at the surface of interfaces between active battery materials will influ-

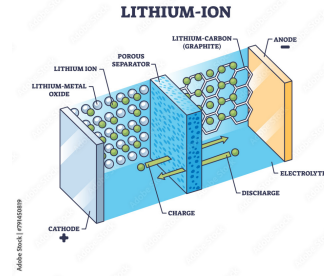


FIG. 1: A standard depiction of lithium ion batteries. This review focuses on phonons in the bulk of a cathode and porous separator, as well as surface interfaces

ence To me it doesn't feel like enough to say it: "will influence". We can already infer that phonons will influence charge transport, but creating models and explanations and predictions for how is more interesting/important. ion transport, charge transfer, and surface reactivity.

II. BACKGROUND

Optical phonon modes involve adjacent atoms moving in opposing directions So, from what I remember from Solid State, there are multiple optical phonon bands that can be created. I know for the two-band case (one acoustic and one optical) that the two different types can be qualitatively described in this way by counting cohesive motion of neighbours versus dis cohesive motion of neighbours. However, I'm not confident that generalizes in such a simple way to every optical phonon band after the first one. Long story short, you might want to preface this by saying "the first one." Or potentially it is absolutely true and the trend continues past the first and I'm just being overly cautious. such that the net momentum is zero even as the unit cell oscillates. Uh, but the phonon itself still has momentum so clearly there is net momentum from something. Phonons are nothing more than normal modes of waves passing through your crystal and because they have a direction they are going

they definitely have momentum. I think there might be a conceptual qualifier I'm missing here when you say it has "no momentum." Energy is stored in the excitation of optical phonon modes which act as sinks, increasing specific heat capacity. Acoustic phonon modes, on the other hand, involve all particles moving in the same direction and are the dominant method for heat dispersion. I don't understand this explanation, but I'm not sure it's particularly important that I do. Personally I am pretty baffled at the idea of heat being localized to a specific position or moving around because I thought it was just a way of keeping track of how the number of accessible states transformed: $dQ \geq TdS$ Feel free to cite some paper that has a more in-depth explanation to appease your audience. The contribution to thermal conductivity k due to the lattice depends on the specific heat capacity of the material C_v , as well as mean free path l , group velocity V_g , and relaxation time τ of acoustic phonons. I think you need to take a lot more time defining all of these concepts. First of all the introduction of group velocity implies we're making the jump from considering isolated phonons stretching across the whole of our material to localized wave packets of phonons. That jump feels like it should be thought out more, especially since a concept like the mean free path is pretty useless for a single phonon since a single phonon wouldn't have a location. I'm not at all familiar with the relaxation time of a phonon, and since you don't seem to mention it elsewhere I think you may as well cut it. Greater mean free path and group velocity yield faster heat dissipated across the solid state lithium ion battery, reducing the critical temperature for thermal runaway. How does more heat equally dispersed decrease the critical temperature for thermal runaway? Wouldn't it just keep the temperature the same but make it a whole lot less likely that we'll ever reach there since the heat is dispersed? This is given by the Debeye Debye equation $k = \frac{1}{3}C_v v l$. You didn't define v which I think is the volume of a unit cell here? Also might be confusing with C_v since you didn't explain that that meant "held at constant volume." This expression doesn't seem to have the group velocity in it, unless I am to understand v as the group velocity?

However, ions traveling through the ion transport layer couple with phonon modes by disrupting the periodicity of the potential in the lattice around it, inducing phonons I am skeptical at the word "inducing". To me induce sounds like you are saying that the ion is creating phonons instead of what it is doing, which is altering your field equations/Hamiltonian perturbatively such that the previous eigenstates (phonons) are no longer eigenstates of the new Hamiltonian and so now have a probability amplitude of transforming into each other because they are no longer stationary states. Everything is Big Quantum by the way, you cannot escape. and causing phonon-phonon collision. This reduces the mean free path, and is the mechanism for electron-phonon coupling. Therefore, there is a tradeoff between the maximum current density and the thermal transport of a lithium ion battery. Do

you have a graph? Visually seeing this tradeoff would be very nice. If you don't actually have a graph with data, you could make a cartoon sketch of one and that would be very helpful! As the temperature increases, the structural order of the crystal decreases, increasing phonon scattering and reducing thermal transport, heating further in a feedback cycle. I don't think I understand the reasoning: hotter-less periodic-more phonon scattering-less thermal transport-hotter. To me it seems like it would go: hotter-less periodic-more phonon scattering-less thermal transport-more localized temperature. There is a minor difference there.

The destructiveness of thermal runaway is given by three temperatures: T_1 , the onset temperature of self-heating, T_2 the TR trigger temperature, and T_3 , the maximum temperature during TR. For best safety, a high T_1 and T_2 is desired, and a low T_3 . I do not understand the difference between T_1 and T_2 here. T_1 seems to be the temperature at which it starts heating itself via exothermic reactions, but I don't see how that heating wouldn't bring it all the way to T_2 and continue to a thermal runaway, and also invalidating the existence of T_2 . I think it would be very helpful to make these three temperatures their own section and explain how measurements can be done to identify them. Do they just heat up their sample and measure the rate at which it heats up as they do so? Also you didn't establish the acronym "TR" earlier. Please do so as it is a useful acronym.

III. PHONONS IN NCM CATHODES

$\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ ternary What does ternary mean? cathodes are highly investigated due to their energy density. However, they seriously suffer from thermal runaway and safety issues, preventing commercial adoption.

Shi et al. investigated electron-phonons coupling in four conventional $\text{Li}[\text{Ni}_x\text{Co}_y\text{Mn}_z]\text{O}_2$ ternary cathode materials: NCM111, NCM523, NCM622, and NCM811 ($x:y:z = 1:1:1, 5:2:3, 6:2:2, 8:1:1$, and $x + y + z = 1$). I believe that these are the ratios of the elements in the unit cells, but I'm not sure. Please explain this notation. Conventionally researchers increase the Ni content in the cathode to reduce the band gap, which shrinks from 0.6 eV to 0.2 eV from NCM111 to NCM811 as the cathode becomes more metal like. Personally I think it's redundant to then specify that the material becomes more metal like, but that might just be me. You should probably still keep this, I just don't like zoology. However, the higher electrical conductivity and carrier density results in stronger electron-phonon interactions, decreasing the mean free path. I don't think you've explained this. It's easy for me to see that floating ions messing up a lattice structure would mess with the mean free path, but I don't understand how the electrons (those tiny little buzzing things with barely any mass at all) would interact with the coherent motion of the nuclei (which would be comparably massive.) The thermal conductiv-

ity drops from 4.5 to 3.5 $W/(m \cdot K)$ from NCM111 to NCM 811. The phonon sound velocity I'm assuming by sound velocity you mean the phase velocity? I'm honestly not sure, and so maybe it would be a benefit to specify. also decreases as the Ni content increases. Combined, these lead to strongly degraded thermal safety. There is a trade-off between electronic conduction and thermal transport properties for cathode materials. Comparing NCM111 to NCM811, it would found that the higher Ni content reduced T_2 of the cell from 223 to 171 C, while T_3 was increases from 782 to 844 C, implying that a higher Ni content significantly deteriorates the thermal stability of the cell. Understanding the relationship between bandgap, electrical conductivity, ion mobility and electron - phonon coupling is required to design high performance but also safe LIBs [1]. So, did they understand it? Or did they just measure it at a couple of different values? If they didn't produce a model then you could try making a linear or quadratic fit of their data and say that you've defined an ad-hoc relationship which you hope can be better built on in the future. If they did present a model or equation relating these various factors, then please do present it.

IV. PHONONS IN SOLID POLYMER ELECTROLYTES

In solid polymer electrolytes(SPE), phonon scattering is mainly driven by chain entanglement, chain ends, bond angle disorder, and chain twisting of the polymers. Those were a whole lot of words that you didn't explain. I would recommend being more thorough if you can with graphical pictures. Or if you don't have the time, then say "polymer chain interactions such as chain entanglement." so there is only one thing your audience is scratching their heads about. Phonon propagation is random when in contact with the irregular potential fields of standard SPEs, reducing thermal conductivity. Most amorphous polymers have thermal conductivity $\kappa = 0.1 - 0.5 WmK^{-1}$. Comparing this with other materials would be very nice since as is I don't know whether this is a high number or not. A small table with Copper, $Li[NiCoMn]O$, and these SPEs would be very useful.

To reduce the scattering, Shi et al [2] introduce an ultrathin polyethylene nanopore separator matrix (PPIL) PPIL doesn't look like the correct acronym. Maybe spell out what each word in PPIL stands for? Also "nanopore separator matrix" is a mouthful and talking through what that means would be useful. I would guess that it is a bunch of polymers tied together with holes on the scale of a nanometer, but I don't know what the word separator corresponds to., Figure 2, wherein the polymer chains regularly pack with minimal kinks. How can they check there are minimal kinks? That sounds difficult. They also introduce an ionic liquid (1-ethyl-3-methylimidazolium dicyanide, EMIM:DCA), polyurethane (PU), and lithium salt on the polyethy-

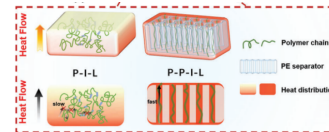


FIG. 2: A depiction of the organizational structure of the PPIL separator matrix, with the solid polymer electrolyte organized along the array for better heat transfer across the separator.

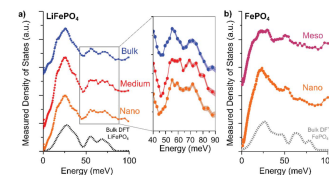


FIG. 3: The phonon density of states for various energies in bulk cathode particles, medium sized particles, and nano particles of lithiated $LiFePO_4$.

lene separator to reduce random intermolecular attractions between the polymer chains. The straight polymer chains lead to more regular thermal transport pathways, thus higher k . Due to the clear pathways for ion transport I thought you just said the PPIL matrix allowed for straight pathways for thermal transport? How/why are they creating pathways for ions as well?, the discharge capacity and cycling performance increased and resistance dropped an order of magnitude with the PPIL electrolyte. Graph? Comparison of data?

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In the PPIL electrolyte, Shi et al found that T_1 increased from 100.5 C in baseline pouch cell to 208.6 C, which was delayed by 6×10^4 s of run time. I don't know what this "delayed by run time" means. Meanwhile, T_2 increases from 160.4 to 234.5 °C, and T_3 disappears from 657.4C[4], implying that the PPIL electrolyte significantly reduced phonon scattering and the conditions for thermal runaway. I'm going to say it again to be repetitive: I think your paper would greatly benefit from a section explaining how to measure T_1 , T_2 , and T_3

I am sorry to say that I do not know what I am meant to take away from this graph. The best I can think to say is that there are more phonons in the bulk of the material which makes sense? I'm not sure how that's relevant. I also don't understand how the DFT calculation's prediction for the Bulk density of states is so off compared to what was actually measured. That rattles my faith in their approach.

V. PHONONS IN LFP CATHODES

Lithium Iron Phosphate cathodes are also commonly explored for high performance lithium ion batteries. Benedek et. all explore surface and bulk phonons and how they couple with ion transport. The speed at which several processes occur depends on or are influenced by local distortions of the crystal lattice of the active material and their vibrational energy. Some examples are ion exchange to or from the electrolyte, electron charge transfer to or from the conducting additives I think conducting additives are those parts where the electrons transfer from the anode to the cathode the long way around generating electricity, but I'm not sure., electrolyte reactions What's an electrolyte reaction and are they something we want? I would have guessed we don't want them. for solid electrolyte interfaces, and catalytic phase changes. These follow many reaction mechanisms, but primarily depend on the local potential fields. Lithium ions can be modeled as hopping across vacancies in lattices, with an energy barrier between each hop. Phonons in LiFePO₄ couple with the activation barriers for lithium ion transport across the boundary. The lattice shifts due to phonons reduce the energy barrier by bringing the lattice sites closer together, which combine with thermal fluctuations to cause ions to hop and transport across the surface. Great and sensible explanation.

A. Interior and Surface vibrational modes for Ion Transport

Surface pPhonons follow different behavior in the bulk of a material compared to the surface, This is a comma splice. I'm pretty sure. Anyway, you don't need a comma here. due to the 2-D nature of the surface restricting the set of allowable modes which don't deflect. The density of states of phonons I am not confident that Engineers and Chemists understand what a density of states is, so you would need to explain this. in the bulk of a LiFePO₄ cathode was explored by measuring inelastic neutron scattering This experimental techniques sounds weird, interesting, and exciting! It's a shame you don't explain it. for various sizes of plates of the LiFePO₄, adjusting the ratio of bulk to surface area of the material for the phonon structure to understand the structure when bulk dominates compared to strong participation from surface modes, as shown in figure 3.

Using inelastic neutron scattering and density functional theory Your audience is meant to be Sophomores who just finished the core, and they probably haven't heard of DFT. So you've got to give a brief explanation about how it's a computational technique to model the electron-based interactions of complex ensembles of atoms, Benedek et al. [3] found that they can control the vibrational energies allowed to surface phonons, by applying carbon surface coatings, giving a way to tune the ion transport by tuning the ways that phonons change

the potential landscape. I don't understand this. I think this deserves at least 2-3 more sentences and an image. Changing the coordination I believe "coordination" is slang for: "number of nearest neighbours", but you might want to be more explicite. and bonding of surface atoms adjusts the vibrational energies at the surface. Yes, but in what way? Does a greater coordination (number of nearest neighbours I believe) increase the density of phonons or decrease it for example?

VI. INTERACTIONS

The amplitudes of phonons I don't think phonons have amplitudes? Each phonon is a discrete quantization of a specific normal mode. You can count up the number of phonons that exist in a single normal mode in order to get at the amplitude of that mode I believe. and which available modes have been seen to both regulate the ways that energy can be dispersed through a crystal, as well as the energy barriers for ion transport at surface sites in cathodes. I feel like that last part was only tangentially touched upon qualitatively. The mean free path of phonons is generally reduced by high ion transport, which could merit further investigation as a self regulating limit to ion transport, and therefore reducing thermal stresses. It has been demonstrated that smooth regular paths (such as the nicely tiled polymer chains at the porous separator) allow for long ranging acoustic phonons to transmit energy across long distances and increase T₁, reducing the dangers of thermal runaway.

The discharge rates of the batteries determine parameterize how much energy is being expelled into heat and lattice vibration in any given time, so the phonon density of states for a design should align to be deep with the desired energy output of the battery to maximize storage and transport of excess energy in non thermal means. [5] This sounds like a very interesting assertion, but I don't really understand it. If you want to workshop this statement feel free to talk it through with me because it feels important.

VII. IMPACT ON SOCIETY

Thermal Runaway is a significant engineering challenge in uses such as electric vehicles and large grid storage, limiting the capabilities of otherwise high capacity lithium ion batteries due to safety risks. While more research is needed to understand the interactions of phonons in solid state batteries, Shi et. al. have conducted a comprehensive investigation of phonons in both the cathode and electrolyte polymer of lithium ion batteries and shown their importance for managing thermal runaway, as well as design considerations for the tradeoff between total current capacity and thermal conductivity due to the largely unexplored mechanism of electron-phonon coupling. The power of phonons

for controlling catalyst chemistry, ion transport, and heat conductivity demonstrates new design considerations and methodology^{ies} to tune performance parameters of lithium ion batteries, such as safety control - opening room for further investigation of safety control in consumer electronics. I don't think you need this extra bit attached to an already longwinded sentence.

ACKNOWLEDGMENTS

Thank you Professors VR and Saeta for offering this class on top of your regular teaching load. Also, my gratitude to Jacob Sesate and Simone Yang for listening to me rant about how cool this was and figure out my thoughts (and also to even parse the papers in the first place). Further thanks to Professors Gerbode and Ritz for inspiring my curiosity in phonons.

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- [1] Shi, X., Jia, Z., Wang, D., Wang, Q., He, D., & Huang, Y. (2023). Achieving High Safety for Lithium-Ion Batteries by Optimizing Electron and Phonon Transport. *ACS Energy Letters*, 8(11), 4540–4546. <https://doi.org/10.1021/acsenergylett.3c01787>
 - [2] Shi, X., Jia, Z., Wang, D., Jiang, B., Liao, Y., Zhang, G., Wang, Q., He, D., & Huang, Y. (2024). Phonon Engineering in Solid Polymer Electrolyte toward High Safety for Solid-State Lithium Batteries. *Advanced Materials*, 36(33), 2405097. <https://doi.org/10.1002/adma.202405097>
 - [3] Benedek, P., Yazdani, N., Chen, H., Wenzler, N., Juranyi, F., Månsson, M., Islam, M. S., Wood, V. C. (2019). Surface phonons of lithium ion battery active materials. *Sustainable Energy and Fuels*, 3(2), 508–513. <https://doi.org/10.1039/c8se00389k>
 - [4] I do not understand what.
 - [5] I don't know if this is actually true, but seems plausible as something to investigate, and its my prospectus so hah.